

Norms and exclusion theorems

By

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The purpose of this paper is to state and prove some theorems which relate the characteristic roots of two matrices.

1. Notation. Let $\|x\|$ denote the norm of a finite dimensional vector x . Let $\|A\|$ denote any norm of a square matrix A satisfying

$$\|Ax\| \leq \|A\| \|x\|.$$

(Such a matrix norm is said to be "consistent with" the vector norm.) Let $\text{lub}(A)$ denote the least upper bound of A , i.e.,

$$\text{lub}(A) = \min_{\alpha} \{ \alpha : \|Ax\| \leq \alpha \|x\| \text{ for all } x \}.$$

It is easy to see that¹

$$\text{lub}(A) \leq \|A\|.$$

For the concepts and notation used here, see HOUSEHOLDER [3] and OSTROWSKI [5].

2. Let x be a characteristic vector of A and let λ be the corresponding characteristic root, i.e., $Ax = \lambda x$. Then, if B is an arbitrary matrix either λ is a characteristic root of B or

$$x = (\lambda I - B)^{-1}(A - B)x.$$

In the second case, we get

$$\|x\| \leq \text{lub}[(\lambda I - B)^{-1}(A - B)] \|x\|$$

and therefore

$$1 \leq \text{lub}[(\lambda I - B)^{-1}(A - B)] \leq \|(\lambda I - B)^{-1}(A - B)\|.$$

Therefore we have

Theorem I. The set $\{z : \|(zI - B)^{-1}(A - B)\| \geq 1 \text{ or } \det(zI - B) = 0\}$ contains all characteristic roots of A .

The multiplicative property of the least upper bound can be used to derive the weaker condition

$$1 \leq \text{lub}[(\lambda I - B)^{-1}] \cdot \text{lub}(A - B) \leq \|(\lambda I - B)^{-1}\| \cdot \|A - B\|.$$

From this follows

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¹ $\text{lub}(A)$ is a norm, too. Moreover, $\text{lub}(AB) \leq \text{lub}(A) \text{lub}(B)$. Thus $\text{lub}(A)$ is a multiplicative norm, although $\|A\|$ may not be.

Theorem II. The set $\{z: \|(zI - B)^{-1}\|^{-1} \leq \|A - B\| \text{ or } \det(zI - B) = 0\}$ contains all characteristic roots of A .

Interesting special cases can be obtained from Theorems I and II by taking particular norms. For example, take $\|x\| = \max |x_i|$ (or $\|x\| = \sum_{i=1}^n |x_i|$) and the corresponding least upper bound

$$\text{lub}(A) = \max_i \sum_{k=1}^n |a_{ik}| \quad \left(\text{or } \text{lub}(A) = \max_k \sum_{i=1}^n |a_{ik}| \right).$$

By taking B to be the diagonal of A , we get in this case the classical theorems of GERSHGORIN [1]. HOUSEHOLDER [2] has already used norms to prove these theorems. More general vector norms can be made to yield the more complicated results of LENG [4].

3. In foregoing sections, no restrictions were made on either A or B . If we make the assumption that B is normalizable, i.e., similar to a diagonal matrix λ we can obtain the further results described below. Let

$$B = P \Lambda P^{-1}.$$

Then

$$\lambda I - B = P(\lambda I - \Lambda) P^{-1}$$

and

$$1 \leq \text{lub}[A(\lambda I - A)^{-1}] \text{lub}(P) \text{lub}(P^{-1}) \text{lub}(A - B) \leq \|G(I - B)^{-1}\| \kappa(P) \|A - B\|,$$

where

$$\kappa(P) = \text{lub}(P) \cdot \text{lub}(P^{-1}).$$

(Note that $\kappa(P) \geq 1$, since for any vector $z \neq 0$

$$z = P P^{-1} z \text{ implies } \|z\| \leq \text{lub}(P) \cdot \text{lub}(P^{-1}) \cdot \|z\|.)$$

We have now

Theorem III. The set $\{z: \|(zI - A)^{-1}\|^{-1} \leq \kappa(P) \|A - B\| \text{ or } \det(zI - B) = 0\}$ contains all characteristic roots of A .

This theorem provides a means of studying the effect on the characteristic roots of small changes in the elements of a matrix. In this context, $A - B$ would probably be regarded as an error matrix. For applications the result leaves something to be desired, since the same bound is given for every characteristic root.

4. We now start from Theorem II again and make use of the spectral decomposition of a matrix B having simple characteristic roots:

$$B = \sum_{k=1}^n \lambda_k E_k.$$

Note that B is normalizable, in this case. The E_k are the principal idempotents (Frobenius covariants) of B . The resolvent of B has the representation

$$(zI - B)^{-1} = \sum_{k=1}^n (z - \lambda_k)^{-1} E_k.$$

Applying Theorem II, we get the set

$$M = \left\{ z: 1 \leq \|A - B\| \sum_{k=1}^n |z - \lambda_k|^{-1} \|E_k\| \text{ or } \det(zI - B) = 0, \right\},$$

which is contained in the union of the n sets

$$M_i = \{z: |z - \lambda_i| \leq n \|A - B\| \cdot \|E_i\|\}.$$

This gives

Theorem IV. The set $M \subset \bigcup_{i=1}^n M_i$ contains all characteristic roots of A .

If $|\lambda_i - \lambda_j| > n \|A - B\| (\|E_i\| + \|E_j\|)$, the intersection $M_i \cap M_j$ is empty. M_i has no intersection with $\bigcup_{\substack{j=1 \\ j \neq i}}^n M_j$ for $\|A - B\|$ sufficiently small, e.g.,

$$(*) \quad \|A - B\| < \frac{1}{n} \min_{k: k \neq i} \frac{|\lambda_i - \lambda_k|}{\|E_i\| + \|E_k\|}.$$

Then M_i contains exactly one characteristic root, since the set M_i degenerates to the point λ_i as $\|A - B\| \rightarrow 0$. Now if $j \neq i$ and $z \in M_i$, then

$$|z - \lambda_j| \geq |\lambda_i - \lambda_j| - n \|A - B\| \cdot \|E_i\|.$$

Applying the condition (*) gives

$$\begin{aligned} |z - \lambda_j| &\geq |\lambda_i - \lambda_j| \cdot \frac{\|E_j\|}{\|E_i\| + \|E_j\|}, \\ \frac{\|E_j\|}{|z - \lambda_j|} &\geq \frac{\|E_i\| + \|E_j\|}{|\lambda_i - \lambda_j|}, \end{aligned}$$

and from the definition of M and Theorem IV

$$1 - \|A - B\| \sum_{j \neq i} \frac{\|E_i\| + \|E_j\|}{|\lambda_i - \lambda_j|} \leq \frac{\|E_i\|}{|z - \lambda_i|} \|A - B\|.$$

Condition (*) makes the left side of this inequality positive. The result can now be expressed as

Theorem V. If $\|A - B\|$ satisfies condition (*), each of the sets

$$M'_i = \left\{ z: |z - \lambda_i| \leq \|E_i\| \cdot \frac{\|A - B\|}{1 - k \|A - B\|} \right\}, \quad \text{where } k = \sum_{j \neq i} \frac{\|E_i\| + \|E_j\|}{|\lambda_i - \lambda_j|}$$

contains exactly one characteristic root of A . This result implies that as $\|A - B\| \rightarrow 0$, the circle around the characteristic root λ_i of B containing a characteristic root of A goes to zero with $\|E_i\| \cdot \|A - B\|$. Similar results, replacing $|z - \lambda_i|$ by $|z - \lambda_i|^v$, hold if λ_i belongs to an elementary divisor of degree v . Theorem V does not suffer from the deficiencies attributed to Theorem III, and it is important to appreciate the significance of this. The circles of Theorem V do not all have the same radii.

5. Theorem III can be simplified if the vector norm belongs to a special class which includes the most used ones — the Hölder norms and diagonal transforms of them. If

$$\text{lub}(D) = \max_i |d_{ii}|,$$

where D is a diagonal matrix, we call $\text{lub}(A)$ axis-oriented. Then for axis oriented norms, we have

Theorem IIIa. The set $\{z: \min_i |z - \lambda_i| \leq \kappa(P) \|A - B\|\}$ or equivalently the union of the circles

$$\{z: |z - \lambda_i| \leq \kappa(P) \|A - B\|\}$$

contains all characteristic roots of A .

6. We now discuss the quantity $\|E_i\|$ which occurs in Theorem IV and Theorem V. Note that $E_i = x_i y_i^H$, where x_i is a right and y_i^H is a left characteristic vector. $\text{Trace}(E_i) = 1$ means $y_i^H x_i = 1$. Now if the vector norm is strictly homogeneous, i.e., if $\|\beta x\| = |\beta| \cdot \|x\|$, then

$$\begin{aligned} \text{lub}(E_i) &= \min \{ \alpha: |y_i^H x| \cdot \|x_i\| \leq \alpha \|x\| \} \\ &= \|x_i\| \cdot \|y_i^H\|_D, \end{aligned}$$

where $\|y_i^H\|_D = \min \{ \alpha: |y_i^H x| < \alpha \|x\| \text{ for all } x \}$ is the dual norm to the given vector norm. Define a cosine of two vectors a, b^H by

$$\cos(a, b^H) = \frac{b^H a}{\|a\| \cdot \|b^H\|_D}.$$

Then

$$\text{lub}(E_i) = \frac{1}{\cos(x_i, y_i^H)}$$

is the reciprocal cosine of the left and right characteristic vectors. Moreover, $\text{lub}(E_i)$ gives for axis-oriented, strictly homogeneous norms a lower bound for $\kappa(P)$, since

$$\text{lub}(E_i) = \|x_i\| \cdot \|y_i^H\|_D = \|P e_i\| \|e_i^H P^{-1}\|_D \leq \kappa(P) \|e_i\| \|e_i^H\|_D,$$

$$\max_i \frac{1}{\cos(x_i, y_i^H)} = \max_i \text{lub}(E_i) \leq \kappa(P).$$

Thus, if $\kappa(P) = 1$, $\text{lub}(E_i) = 1$ or x_i and y_i^H are dual vectors. We call a matrix B normal if this is true for all pairs of characteristic vectors, and we can consider $\max_i \text{lub}(E_i)$ a measure of nonnormality of a normalizable matrix, and $\kappa(P)$ an indication of this measure.

For euclidean norm, the inequality of KANTOROVICH [6] gives even the sharper result

$$\frac{\|P e_i\| \|e_i^H P^{-1}\|}{\|e_i\| \|e_i^H\|} \leq \frac{\sigma_{\max}^2(P) + \sigma_{\min}^2(P)}{2 \sigma_{\max}(P) \sigma_{\min}(P)}$$

where $\sigma_{\max}(P) = \text{lub}(P)$, $\sigma_{\min}(P) = [\text{lub}(P^{-1})]^{-1}$ the largest and smallest singular value, resp.

Therefore, for euclidean norm

$$\max_i \frac{1}{\cos(x_i, y_i^H)} = \max_i \text{lub}(E_i) \leq \frac{1}{2} \left(\kappa(P) + \frac{1}{\kappa(P)} \right).$$

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